SE MID PRACTICE QUESTIONS 40 MCQS --------LECTURE 02----------

1. What is the primary purpose of the Software Development Life Cycle (SDLC)?

A) To manage the project budget

B) To describe the main phases involved in creating a software product

C) To specify the programming languages to be used

D) To document user feedback

E) To replace the need for a software process model

Answer: B) To describe the main phases involved in creating a software product

Explanation: Slide 3 defines SDLC as a "step-by-step process" that "describes the main phases involved in creating a software product."

2. A Software Process Model is best described as:

A) A specific programming paradigm like object-oriented programming

B) A structured method that guides how the SDLC phases are carried out

C) A tool for version control

D) A type of software testing

E) The final product delivered to the customer

Answer: B) A structured method that guides how the SDLC phases are carried out

Explanation: Slide 3 defines a Software Process Model as a "structured method or strategy that guides how the SDLC phases are carried out."

3. Which of the following is an example of a Software Process Model?

A) Java

B) MySQL

C) Waterfall

D) GitHub

E) Python

Answer: C) Waterfall

Explanation: Slide 3 lists Waterfall, Spiral, and Agile as examples of Software Process Models. The others are programming languages, databases, or tools.

4. How does a Software Process Model relate to the SDLC?

A) It is another name for the SDLC.

B) It defines how to execute the phases defined by the SDLC.

C) It is only used after the SDLC is complete.

D) It replaces the need for the SDLC in modern projects.

E) It is unrelated to the SDLC.

Answer: B) It defines how to execute the phases defined by the SDLC.

Explanation: Slide 4 clarifies the connection: SDLC "defines what phases to follow," while a model "defines how to execute those phases."

5. The Waterfall Model is characterized as:

A) Iterative and flexible

B) Linear and phase-based

C) Risk-driven and cyclical

D) Based on early prototyping

E) Focusing on continuous delivery

Answer: B) Linear and phase-based

Explanation: Slide 6 describes the Waterfall Model as a "linear and step-by-step process."

6. In which situation is the Waterfall Model most appropriate?

A) When requirements are unclear and likely to change

B) For a research and development project

C) When project requirements are stable and well-defined

D) When the client demands daily deliverables

E) For a high-risk, complex system

Answer: C) When project requirements are stable and well-defined

Explanation: Slide 8 states the Waterfall Model is best when "requirements are clearly known in advance and unlikely to change."

7. The key improvement of the Iterative Waterfall Model over the classic Waterfall Model is that it:

A) Eliminates the testing phase

B) Allows teams to go back to earlier phases for corrections

C) Does not require a design phase

D) Is only used for very small projects

E) Guarantees no risks

Answer: B) Allows teams to go back to earlier phases for corrections

Explanation: Slide 9 states that the Iterative Waterfall Model allows teams to "go back to earlier phases if needed" for fixes and improvements.

SE MID PRACTICE QUESTIONS 40 MCQS --------LECTURE 02----------

8. The Iterative Waterfall Model is best suited for:

A) Projects with completely unknown requirements

B) Safety-critical systems like avionics software

C) Medium-sized systems where requirements are mostly clear but minor changes may be needed

D) Projects requiring immediate, continuous delivery

E) Very large projects with high levels of risk

Answer: C) Medium-sized systems where requirements are mostly clear but minor changes may be needed

Explanation: Slide 12 lists these conditions as suitable for the Iterative Waterfall Model.

9. The V-Model emphasizes:

A) Speeding up the coding phase

B) Reducing customer collaboration

C) Testing and validation at every stage of development

D) Using only a single, final test at the end

E) Avoiding documentation

Answer: C) Testing and validation at every stage of development

Explanation: Slide 14 states the V-Model "focuses on testing at every stage of development."

10. On the "V" in the V-Model, where does development happen?

A) On the right side

B) On the left side

C) At the bottom

D) At the top

E) It is not represented on the "V"

Answer: B) On the left side

Explanation: Slide 14 states that "development happens on the left side of the 'V'".

11. The V-Model is considered ideal for:

A) Social media applications

B) Safety-critical systems like medical devices

C) Rapidly developed startup MVPs

D) Projects with highly volatile requirements

E) Systems where late changes are expected

Answer: B) Safety-critical systems like medical devices

Explanation: Slide 15 lists "medical devices" and "automotive systems" as ideal use cases for the V-Model due to its rigorous testing approach.

12. The primary goal of the Prototyping Model is to:

A) Deliver the final product in the first iteration

B) Build a quick, rough version to gather user feedback and refine requirements

C) Eliminate the need for a development team

D) Follow a plan-driven approach without feedback

E) Create comprehensive documentation before any coding

Answer: B) Build a quick, rough version to gather user feedback and refine requirements

Explanation: Slide 17 describes the model's process: build a quick prototype, show it to the user, get feedback, and improve it.

13. The Prototyping Model is particularly useful for projects where:

A) Requirements are unclear or may change

B) The budget is extremely fixed and cannot change

C) The deadline is years away

D) Requirements are stable and well-defined

E) No user interaction is involved

Answer: A) Requirements are unclear or may change

Explanation: Slide 18 lists this as a key reason to use the Prototyping Model.

14. In the Incremental Model, the software is developed and delivered as:

A) One complete system at the end

B) Small parts or increments

C) A single, throwaway prototype

D) A complete set of documents first

E) A risk analysis report

Answer: B) Small parts or increments

Explanation: Slide 19 defines the Incremental Model as developing the system "step by step in small parts called increments."

SE MID PRACTICE QUESTIONS 40 MCQS --------LECTURE 02----------

15. Each increment in the Incremental Model typically goes through:

A) Only the implementation phase

B) The entire SDLC

C) Only the testing phase

D) Only the requirements phase

E) A single spiral loop

Answer: B) The entire SDLC

Explanation: Slide 19 explicitly states that "every increment goes through the entire SDLC."

16. A major advantage of the Incremental Model is that it allows for:

A) Early delivery of useful features and ongoing improvements

B) Hiding the project progress from the customer

C) Avoiding any planning or design

D) Completely eliminating project risk

E) Finalizing all requirements before any work begins

Answer: A) Early delivery of useful features and ongoing improvements

Explanation: Slide 19 cites this as the "Why use it?" reason for the model.

17. The Spiral Model combines features of which two other models?

A) Waterfall and Prototyping

B) V-Model and Agile

C) Incremental and Scrum

D) Waterfall and V-Model

E) Kanban and Prototyping

Answer: A) Waterfall and Prototyping

Explanation: Slide 22 states the Spiral Model "combines features of both the Waterfall Model and Prototyping Model."

18. What is the defining characteristic of each loop in the Spiral Model?

A) It must be completed in less than one week.

B) It includes a phase dedicated explicitly to risk analysis.

C) It delivers the final, shippable product.

D) It does not require customer evaluation.

E) It avoids planning until the very end.

Answer: B) It includes a phase dedicated explicitly to risk analysis.

Explanation: Slide 23 lists "Risk Analysis" as one of the four fundamental phases of every spiral loop, which is its key differentiator.

19. The Spiral Model is most appropriate for:

A) Small, simple projects with low risk

B) Large, complex, or high-risk projects

C) Projects where requirements will not change

D) Projects that need a single, final delivery

E) Maintaining legacy systems with no new features

Answer: B) Large, complex, or high-risk projects

Explanation: Slide 24 lists these as the conditions for using the Spiral Model.

20. The Agile Model focuses on: (Select all that apply)

A) Team collaboration and customer feedback

B) Flexibility to change

C) Delivering software in small, usable parts

D) Extensive documentation before any coding

E) Following a rigid, unchangeable plan

Answer: A) Team collaboration and customer feedback, B) Flexibility to change, C) Delivering software in small, usable parts

Explanation: Slide 26 lists these as the focus of the Agile model. Options D and E are characteristics of plan-driven models, not Agile.

21. In Agile, a short, time-boxed iteration of work is commonly called a:

A) Phase

B) Loop

C) Spiral

D) Sprint

E) Cycle

Answer: D) Sprint

Explanation: Slide 26 states that in Agile, software is built and delivered in "small usable parts called sprints."

SE MID PRACTICE QUESTIONS 40 MCQS --------LECTURE 02----------

22. Agile is based on the values and principles expressed in the:

A) IEEE Standards

B) Agile Manifesto

C) SWEBOK Guide

D) Waterfall Protocol

E) Spiral Contract

Answer: B) Agile Manifesto

Explanation: Slide 26 states Agile is "based on the Agile Manifesto (2001)."

23. According to the lecture, Agile is best described as a:

A) Rigid set of rules

B) Philosophy that is implemented using frameworks

C) Single, specific methodology

D) Type of programming language

E) Replacement for all other models

Answer: B) Philosophy that is implemented using frameworks

Explanation: Slide 28 explicitly states "Agile is not one specific method. It is a philosophy" that is applied using frameworks like Scrum and Kanban.

24. Which of the following are popular frameworks used to implement Agile? (Select all that apply)

A) Waterfall

B) Scrum

C) V-Model

D) Kanban

E) Spiral

Answer: B) Scrum, D) Kanban

Explanation: Slide 28 lists Scrum and Kanban as the "two most popular Agile frameworks."

25. According to the comparison table (Slide 30), which model is best for safety-critical systems like avionics?

A) Waterfall

B) Prototyping

C) V-Model

D) Incremental

E) Agile

Answer: C) V-Model

Explanation: The table on Slide 30 lists "Safety-critical systems (e.g. avionics)" in the "Best For" column for the V-Model.

26. According to the comparison table, a key drawback of the Prototyping Model is:

A) No feedback until the end

B) The prototype may be mistaken as the final product

C) Complex and expensive

D) Rigid schedules and roles

E) Requires stable architecture upfront

Answer: B) The prototype may be mistaken as the final product

Explanation: Slide 30's table lists "Prototype may be mistaken as final; scope creep" as a drawback of the Prototyping Model.

27. Which model is described as "Risk-driven" in its approach?

A) Waterfall

B) Incremental

C) Spiral

D) Scrum

E) Kanban

Answer: C) Spiral

Explanation: The table on Slide 30 lists the "Approach" of the Spiral Model as "Risk-driven."

28. When choosing a model, if a project has high risk levels, the lecture suggests considering the:

A) Waterfall Model

B) Spiral Model

C) Prototyping Model

D) V-Model

E) Kanban

Answer: B) Spiral Model

Explanation: Slide 31, in the "How to Choose a Model?" table, states "High-risk projects need risk assessment → Spiral is ideal."

SE MID PRACTICE QUESTIONS 40 MCQS --------LECTURE 02----------

29. If the goal is the fastest possible delivery of working features, the best choice would likely be:

A) Waterfall

B) V-Model

C) Agile/Scrum or Incremental

D) Spiral

E) Iterative Waterfall

Answer: C) Agile/Scrum or Incremental

Explanation: Slide 31 states that if you "Need fast delivery?" you should "Choose Agile, Scrum, or Incremental."

30. The factor "Fixed vs. Changing Requirements" matters when choosing a model because:

A) It determines the programming language.

B) Stable requirements suit Waterfall; changing needs call for Agile or Spiral.

C) It is only relevant for the maintenance phase.

D) It defines the team's salary.

E) It is the only factor that matters.

Answer: B) Stable requirements suit Waterfall; changing needs call for Agile or Spiral.

Explanation: This is a direct quote from the first row of the table on Slide 31.

31. Which model executes the SDLC in a strict, sequential manner?

A) Agile

B) Spiral

C) Waterfall

D) Prototyping

E) Incremental

Answer: C) Waterfall

Explanation: Slide 4 uses the Waterfall model as the example of a model that "executes SDLC sequentially."

32. The "Evaluation" phase in the Spiral Model involves:

A) Writing code for the entire system

B) Customer review of the work done to plan the next iteration

C) Finalizing the budget for the entire project

D) Halting the project indefinitely

E) Ignoring all previous risks

Answer: B) Customer review of the work done to plan the next iteration

Explanation: Slide 23 describes the "Evaluation" phase as where the "Customer reviews the work done" and the "next loop (iteration) is planned."

33. The Windows OS example was used to illustrate which model?

A) Waterfall Model

B) Prototyping Model

C) Incremental Model

D) Spiral Model

E) V-Model

Answer: C) Incremental Model

Explanation: Slides 20 and 21 use the example of Windows 7, 8, 10, and 11 to illustrate delivery via increments.

34. Which model is specifically designed to "catch errors early" by pairing development and testing phases?

A) Prototyping

B) Waterfall

C) V-Model

D) Agile

E) Kanban

Answer: C) V-Model

Explanation: Slide 15 states the V-Model "helps catch errors early by planning tests as soon as each phase begins."

35. In the Iterative Waterfall Model, feedback and revisions are allowed:

A) Only before the project starts

B) Only after the final product is delivered

C) Between phases

D) Never; it is strictly linear

E) Only during the testing phase

Answer: C) Between phases

Explanation: Slide 9 states the Iterative Waterfall Model "allows feedback and revisions between phases."

36. The Agile principle of valuing "individuals and interactions over processes and tools" comes from the:

A) Waterfall Manifesto

B) Spiral Guidelines

C) Agile Manifesto

D) IEEE Standards

E) V-Model Protocol

Answer: C) Agile Manifesto

Explanation: While not explicitly listed in the slides, this is a core value of the Agile Manifesto, which Slide 26 establishes as the basis for Agile.

37. For a project where the client is unsure of their precise needs and needs to see a mock-up first, the most suitable model would be the:

A) Waterfall Model

B) V-Model

C) Prototyping Model

D) Spiral Model

E) Incremental Model

Answer: C) Prototyping Model

Explanation: Slide 18 states the Prototyping Model is "great for projects where requirements are unclear."

38. A key benefit of the Incremental Model is that it helps to:

A) Avoid any interaction with the user until the end

B) Spread out the risk across multiple stages of development

C) Eliminate the need for a design phase

D) Ensure the entire system is perfect before release

E) Hide cost overruns

Answer: B) Spread out the risk across multiple stages of development

Explanation: Slide 20 lists this as a reason to use the Incremental Model.

39. The Scrum framework includes which of the following roles?

A) CEO

B) Project Manager

C) Product Owner

D) System Architect

E) Chief Risk Officer

Answer: C) Product Owner

Explanation: Slide 29 lists "Product Owner" as one of the key roles in the Scrum framework.

SE MID PRACTICE QUESTIONS 40 MCQS --------LECTURE 02----------

40. The activity of building a "quick and rough version" of the system is central to which model?

A) Waterfall

B) V-Model

C) Prototyping

D) Spiral

E) Incremental

Answer: C) Prototyping

Explanation: This is the literal definition of the Prototyping Model provided on Slide 17